

Program 1: Working with Maven: Creating a Maven Project, Understanding the POM File

Dependency Management and Plugins

Step 1: Create a Maven project by running following command into terminal:

```
mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
```

```
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine: $ cd devops/
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine: ~/devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO]
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO]
[INFO] Parameter: basedir, Value: /home/shravan-hegde/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenApp
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/shravan-hegde/devops/MyMavenApp
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 1.053 s
[INFO] Finished at: 2026-03-17T23:58:28+05:30
[INFO]
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine: ~/devops$
```

This command generates a basic Maven project structure with a sample App.java and AppTest.java.

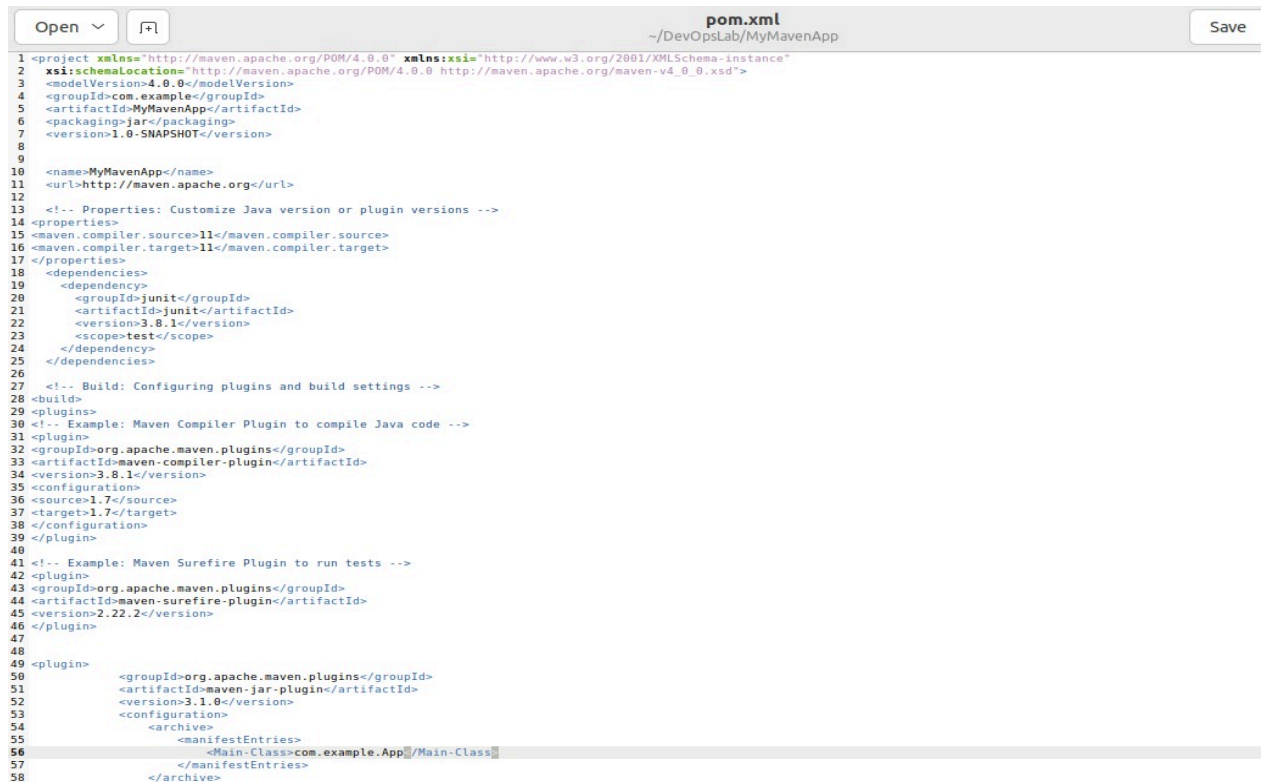
```
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenApp
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/shravan-hegde/devops/MyMavenApp
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 1.053 s
[INFO] Finished at: 2026-03-17T23:58:28+05:30
[INFO]
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine: ~/devops$ tree
.
├── MyMavenApp
│   ├── pom.xml
│   └── src
│       ├── main
│       │   ├── java
│       │   │   ├── com
│       │   │   │   ├── example
│       │   │   │   └── App.java
│       │   └── test
│       │       ├── java
│       │       │   ├── com
│       │       │   │   ├── example
│       │       │   └── AppTest.java
│       └── resources
└── 11 directories, 3 files
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine: ~/devops$
```

This “Build Success” will confirm that the project is created successfully

Step 2: Go to the project directory using command cd MyMavenApp

Step 3: Open and edit pom.xml file using command gedit pom.xml

- Go to <https://mvnrepository.com>
- Search for a dependency “junit”
- Select the latest version and copy the dependency XML snippet.
- Paste it under the <dependencies> section in pom.xml
- Updated pom.xml:



```

1 <?xml version="1.0" encoding="UTF-8"?>
2 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3   xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
4   <modelVersion>4.0.0</modelVersion>
5   <groupId>com.example</groupId>
6   <artifactId>MyMavenApp</artifactId>
7   <packaging>jar</packaging>
8   <version>1.0-SNAPSHOT</version>
9
10  <name>MyMavenApp</name>
11  <url>http://maven.apache.org</url>
12
13  <!-- Properties: Customize Java version or plugin versions -->
14  <properties>
15    <maven.compiler.source>11</maven.compiler.source>
16    <maven.compiler.target>11</maven.compiler.target>
17  </properties>
18  <dependencies>
19    <dependency>
20      <groupId>junit</groupId>
21      <artifactId>junit</artifactId>
22      <version>3.8.1</version>
23      <scope>test</scope>
24    </dependency>
25  </dependencies>
26
27  <!-- Build: Configuring plugins and build settings -->
28  <build>
29    <plugins>
30      <!-- Example: Maven Compiler Plugin to compile Java code -->
31      <plugin>
32        <groupId>org.apache.maven.plugins</groupId>
33        <artifactId>maven-compiler-plugin</artifactId>
34        <version>3.8.1</version>
35        <configuration>
36          <source>1.7</source>
37          <target>1.7</target>
38        </configuration>
39      </plugin>
40
41      <!-- Example: Maven Surefire Plugin to run tests -->
42      <plugin>
43        <groupId>org.apache.maven.plugins</groupId>
44        <artifactId>maven-surefire-plugin</artifactId>
45        <version>2.22.2</version>
46      </plugin>
47
48      <plugin>
49        <groupId>org.apache.maven.plugins</groupId>
50        <artifactId>maven-jar-plugin</artifactId>
51        <version>3.1.0</version>
52        <configuration>
53          <archive>
54            <manifestEntries>
55              <Main-Class>com.example.App</Main-Class>
56            </manifestEntries>
57          </archive>
58        </configuration>
59      </plugin>
60    </plugins>
61  </build>
62 </project>

```

Update App.java:

```

1 package com.example;
2
3 /**
4  * Hello world!
5  *
6  */
7 public class App
8 {
9     public static void main( String[] args )
10    {
11        System.out.println( "Hello World!" );
12    }
13 }

```

Step 4: Build the project using command mvn compile

The project building will start like this.

```

shavan-hegde@shavan-hegde-QEMU-Virtual-Machine:~/Devops/MyMavenApp$ mvn compile
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:MyMavenApp >-----
[INFO] Building MyMavenApp 1.0-SNAPSHOT
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/shavan-hegde/Devops/MyMavenApp/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenApp ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 2 source files to /home/shavan-hegde/Devops/MyMavenApp/target/classes
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 0.778 s
[INFO] Finished at: 2026-03-18T00:15:00+05:30
[INFO]
shavan-hegde@shavan-hegde-QEMU-Virtual-Machine:~/Devops/MyMavenApp$

```

The “Build Success” message ensures that our Maven project is built successfully.

Step 5: Package the project using command mvn package.

```

[INFO] Finished at: 2026-03-18T00:15:00+05:30
[INFO]
shavan-hegde@shavan-hegde-QEMU-Virtual-Machine:~/Devops/MyMavenApp$ mvn package
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:MyMavenApp >-----
[INFO] Building MyMavenApp 1.0-SNAPSHOT
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/shavan-hegde/Devops/MyMavenApp/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MyMavenApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/shavan-hegde/Devops/MyMavenApp/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:testCompile (default-testCompile) @ MyMavenApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- maven-surefire-plugin:2.22.2:test (default-test) @ MyMavenApp ---
[INFO]
[INFO] T E S T S
[INFO]
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.006 s - in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] --- maven-jar-plugin:3.1.0:jar (default-jar) @ MyMavenApp ---

```

Step 6: Run the test on project using command mvn test.

```

shavan-hegde@shavan-hegde-QEMU-Virtual-Machine:~/Devops/MyMavenApp$ mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:MyMavenApp >-----
[INFO] Building MyMavenApp 1.0-SNAPSHOT
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/shavan-hegde/Devops/MyMavenApp/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MyMavenApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/shavan-hegde/Devops/MyMavenApp/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:testCompile (default-testCompile) @ MyMavenApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- maven-surefire-plugin:2.22.2:test (default-test) @ MyMavenApp ---
[INFO]
[INFO] T E S T S
[INFO]
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.008 s - in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] BUILD SUCCESS

```

Step 7: To run the Maven application

- Go to target directory present in MyMavenApp using command `cd target`
- Execute the JAR file with the command:

`java -jar MyMavenApp-1.0-SNAPSHOT.jar`
- After completing the execution, the output will look like:

```
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.008 s - in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.029 s
[INFO] Finished at: 2026-03-18T00:16:43+05:30
[INFO] -----
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyMavenApp$ java -jar target/MyMavenApp-1.0-SNAPSHOT.jar
Hello, World!
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyMavenApp$
```

This will show output as the code present in App.java file we can edit the text to get any other output.

Creating a Maven Project with Guava and Commons-IO

Google Guava Dependency

Guava is a set of core libraries developed by Google to fill gaps in Java's standard libraries. It is used for:

- Collections enhancements: Multimap, BiMap, Table, ImmutableList, etc.
- Functional programming utilities: Predicates, Functions, Suppliers.
- Caching: Built-in caching mechanisms (CacheBuilder) like a mini in-memory cache.
- String utilities: Splitter, Joiner, CharMatcher.
- Concurrency: ListenableFuture, Service, and other helper classes.
- Hashing and hashing strategies: Useful for consistent hashing (e.g., Bloom filters).

Overall, Guava simplifies and improves many core Java patterns and utilities.

Apache Commons IO Dependency

Apache Commons IO is focused purely on input/output utilities, especially for file and stream handling. It is used for:

- File operations: Easily read/write files with FileUtils.
- Stream utilities: IOUtils.copy(), IOUtils.toString() — simplifies stream handling.
- Directory monitoring: FileAlterationObserver for detecting changes in a folder.
- File filtering: WildcardFileFilter, SuffixFileFilter.

Overall, Commons IO reduces boilerplate and errors when working with files or streams.

Project 2:-Maven Project with Guava and Commons-IO Dependencies

Create a Maven Project Using Terminal by following command:

`mvn archetype: generate -DgroupId=com.example -DartifactId=MyMavenGuavaApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false`

```
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~$ cd devops/
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 30 MB/s)
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO]
[INFO] -----
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] -----
[INFO] Parameter: basedir, Value: /home/shravan-hegde/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenApp
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/shravan-hegde/devops/MyMavenApp
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] -----
[INFO] Total time: 1.556 s
[INFO] Finished at: 2026-03-18T00:25:31+05:30
[INFO]
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/devops$
```

Now Navigate to the Project Directory using `cd MyMavenGuavaApp`

Then, use `tree` command to check the directory structure created by our command

```
[INFO] project created from Old (1.x) Archetype in dir: /home/shravan-hegde/devops/MyMavenApp
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.556 s
[INFO] Finished at: 2026-03-18T00:25:31+05:30
[INFO] -----
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/devops$ tree
.
├── MyMavenApp
│   ├── pom.xml
│   └── src
│       ├── main
│       │   ├── java
│       │   │   ├── com
│       │   │   │   └── example
│       │   │   │       └── App.java
│       └── test
│           ├── java
│           │   ├── com
│           │   │   └── example
│           │   │       └── AppTest.java
└── 11 directories, 3 files
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/devops$
```

Edit the pom.xml File to Add Dependencies using gedit pom.xml and inside the

```

1 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
2   xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
3   <modelVersion>4.0.0</modelVersion>
4   <groupId>com.example</groupId>
5   <artifactId>MyMavenApp</artifactId>
6   <packaging>jar</packaging>
7   <version>1.0-SNAPSHOT</version>
8
9
10  <name>MyMavenApp</name>
11  <url>http://maven.apache.org</url>
12
13  <!-- Properties: Customize Java version or plugin versions -->
14  <properties>
15    <maven.compiler.source>11</maven.compiler.source>
16    <maven.compiler.target>11</maven.compiler.target>
17  </properties>
18  <dependencies>
19    <dependency>
20      <groupId>junit</groupId>
21      <artifactId>junit</artifactId>
22      <version>3.8.1</version>
23      <scope>test</scope>
24    </dependency>
25  </dependencies>
26
27  <!-- Build: Configuring plugins and build settings -->
28  <build>
29    <plugins>
30      <!-- Example: Maven Compiler Plugin to compile Java code -->
31      <plugin>
32        <groupId>org.apache.maven.plugins</groupId>
33        <artifactId>maven-compiler-plugin</artifactId>
34        <version>3.8.1</version>
35        <configuration>
36          <source>1.7</source>
37          <target>1.7</target>
38        </configuration>
39      </plugin>
40
41      <!-- Example: Maven Surefire Plugin to run tests -->
42      <plugin>
43        <groupId>org.apache.maven.plugins</groupId>
44        <artifactId>maven-surefire-plugin</artifactId>
45        <version>2.22.2</version>
46      </plugin>
47
48      <plugin>
49        <groupId>org.apache.maven.plugins</groupId>
50        <artifactId>maven-jar-plugin</artifactId>
51        <version>3.1.0</version>
52        <configuration>
53          <archive>
54            <manifestEntries>
55              <Main-Class>com.example.App</Main-Class>
56            </manifestEntries>
57          </archive>
58        </configuration>
59      </plugin>
60    </plugins>
61  </build>
62  </project>

```

Modify App.java

Got to App.java using `cd src/main/java/com/example/` and `gedit App.java` to type the file with following code


```

1 package com.example;
2
3 // Import FileUtils class from Apache Commons IO library
4 // This is used to perform file operations like copying files
5 import org.apache.commons.io.FileUtils;
6
7 // Import File class to represent file and directory pathnames
8 import java.io.File;
9
10 // Import IOException to handle input-output errors
11 import java.io.IOException;
12
13 // Import ImmutableList from Google Guava library
14 // ImmutableList creates a list that cannot be modified after creation
15 import com.google.common.collect.ImmutableList;
16
17 // Define the main class
18 public class App
19 {
20     // Main method - program execution starts from here
21     public static void main( String[] args )
22     {
23         // Create an immutable list of strings (fruits)
24         // This list cannot be changed (no add/remove allowed)
25         ImmutableList<String> fruits = ImmutableList.of("Apple", "Banana", "Cherry");
26
27         // Print the list of fruits to the console
28         System.out.println(fruits);
29
30         // Create a File object representing the source file
31         File sourceFile = new File("source.txt");
32
33         // Create a File object representing the destination file
34         File destFile = new File("destination.txt");
35
36         try {
37             // Copy the contents of source.txt to destination.txt
38             // Using Apache Commons IO FileUtils class
39             FileUtils.copyFile(sourceFile, destFile);
40
41             // Print success message if file is copied
42             System.out.println("File copied successfully!");
43         }
44         catch (IOException e) {
45             // If any error occurs during file copying,
46             // this block will handle the exception
47
48             // Print error message
49             System.err.println("Error occurred while copying file: " + e.getMessage());
50         }
51     }
52 }

```

Build the Project using the command mvn compile command

```

shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyMavenGuavaApp$ gedit pom.xml
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyMavenGuavaApp$ mvn compile
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:MyMavenGuavaApp >-----
[INFO] Building MyMavenGuavaApp 1.0-SNAPSHOT
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenGuavaApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/shravan-hegde/Devops/MyMavenGuavaApp/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenGuavaApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 0.354 s
[INFO] Finished at: 2026-03-18T00:29:58+05:30
[INFO]
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyMavenGuavaApp$

```

```
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyMavenGuavaApp$ mvn package
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:MyMavenGuavaApp >-----
[INFO] Building MyMavenGuavaApp 1.0-SNAPSHOT
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenGuavaApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/shravan-hegde/Devops/MyMavenGuavaApp/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenGuavaApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MyMavenGuavaApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/shravan-hegde/Devops/MyMavenGuavaApp/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:testCompile (default-testCompile) @ MyMavenGuavaApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- maven-surefire-plugin:2.22.2:test (default-test) @ MyMavenGuavaApp ---
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.002 s - in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] --- maven-jar-plugin:3.1.0:jar (default-jar) @ MyMavenGuavaApp ---
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
```

```
[INFO] -----
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyMavenGuavaApp$ mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:MyMavenGuavaApp >-----
[INFO] Building MyMavenGuavaApp 1.0-SNAPSHOT
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenGuavaApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/shravan-hegde/Devops/MyMavenGuavaApp/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenGuavaApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MyMavenGuavaApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/shravan-hegde/Devops/MyMavenGuavaApp/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:testCompile (default-testCompile) @ MyMavenGuavaApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- maven-surefire-plugin:2.22.2:test (default-test) @ MyMavenGuavaApp ---
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.004 s - in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
```

This compiles the project and creates a JAR in the target/ folder.

Run the Application:

If java-jar command give an error which is most probable, use the following command

```
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyMavenGuavaApp$ java -jar target/MyMavenGuavaApp-1.0-SNAPSHOT.jar
Exception in thread "main" java.lang.NoClassDefFoundError: com/google/common/collect/ImmutableList
    at com.example.App.main(App.java:29)
    Caused by: java.lang.ClassNotFoundException: com.google.common.collect.ImmutableList
    at java.base/jdk.internal.loader.BuiltinClassLoader.loadClass(BuiltinClassLoader.java:641)
    at java.base/jdk.internal.loader.ClassLoaders$AppClassLoader.loadClass(ClassLoaders.java:188)
    at java.base/java.lang.ClassLoader.loadClass(ClassLoader.java:525)
    ... 1 more
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyMavenGuavaApp$
```

mvn exec:java -Dexec.mainClass="com.example.App"

This command gives error as

Error while copying file:Source 'source.txt' does not exist

Now, Create a Sample File to Copy using echo "My Data needs to be transferred" > source.txt



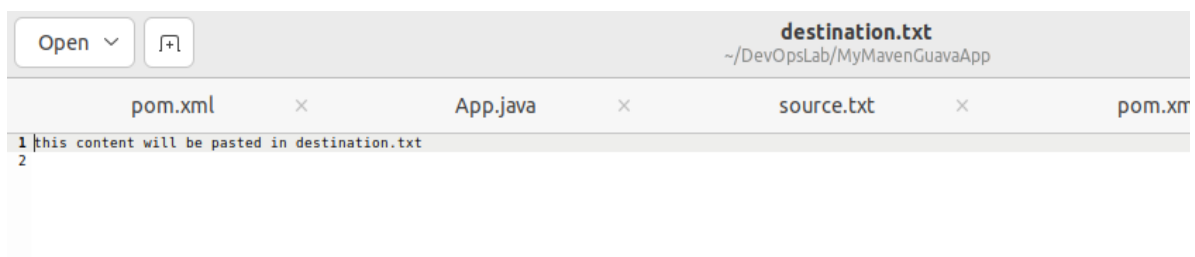
The screenshot shows a code editor window with a tab titled 'source.txt' at the path '~/DevOpsLab/MyMavenGuavaApp'. The editor contains two lines of text: '1 |this content will be pasted in destination.txt' and '2 |'.

Now Run the command `mvn exec:java -Dexec.mainClass="com.example.App"` again

```
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/DevOps/MyMavenGuavaApp$ mvn exec:java -Dexec.mainClass="com.example.App"
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:MyMavenGuavaApp -----
[INFO] Building MyMavenGuavaApp 1.0-SNAPSHOT
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- exec-maven-plugin:3.6.3:java (default-cli) @ MyMavenGuavaApp ---
[Apple, Banana, Cherry]
File copied successfully!
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 0.296 s
[INFO] Finished at: 2026-03-18T00:34:47+05:30
[INFO]
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/DevOps/MyMavenGuavaApp$
```

The output shows file copied successfully.

We can see newly created destination.txt using gedit destination.txt if you are still in Project's directory



The screenshot shows a code editor window with a tab titled 'destination.txt' at the path '~/DevOpsLab/MyMavenGuavaApp'. The editor contains two lines of text: '1 |this content will be pasted in destination.txt' and '2 |'. The editor also shows other open files: 'pom.xml', 'App.java', 'source.txt', and 'pom.xml'.

Project 3:-Maven Selenium Project – Automating Product Selection Using Selenium Objective

To create a **Maven-based Selenium project** that automatically opens a web page, adds selected products to the cart, and views the cart using **Java Selenium WebDriver**.

Sample Website Used for Automation

Websites used for testing:

<https://www.saucedemo.com/>

<https://practicetestautomation.com/practice-test-login/>

<https://automationexercise.com/>

This website contains multiple products that can be added to the cart using automation.

Step 1: Create a Maven Project Using Terminal

Open the terminal and run the following command to create a Maven project.

```
mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp01
-DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
```

This command creates a Maven project named **MyMavenSeleniumApp01**.

```
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~$ cd devops/
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp01
-DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO]
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO]
[INFO] Parameter: basedir, Value: /home/shravan-hegde/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenSeleniumApp01
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/shravan-hegde/devops/MyMavenSeleniumApp01
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] -----
[INFO] Total time: 0.781 s
[INFO] Finished at: 2026-03-18T01:00:45+05:30
[INFO]
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/devops$
```

Project Structure:

```
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/devops/MyMavenSeleniumApp01$ tree
.
├── pom.xml
├── src
│   ├── main
│   │   ├── java
│   │   │   ├── com
│   │   │   │   ├── example
│   │   │   │   │   └── App.java
│   │   └── test
│   │       ├── java
│   │       │   ├── com
│   │       │   │   ├── example
│   │       │   │   │   └── AppTest.java
│   │       └── resources
│   │           └── log4j.properties
└── test
```

10 directories, 3 files
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/devops/MyMavenSeleniumApp01\$

Step 2: Install ChromeDriver

Before running Selenium scripts, ChromeDriver must be installed.

Check Installed Chrome Version

Run the following command:

```
google-chrome --version
```

Example Output:

Google Chrome 133.0.6943.141

Extract Chrome Version

```
CHROME_VERSION=$(google-chrome --version | awk '{print $3}' | cut -d'.' -f1-3)
```

This extracts the Chrome version and stores it in the variable **CHROME_VERSION**.

Install jq (Used for JSON Parsing)

```
sudo apt install jq
```

Install curl (If Not Installed)

```
sudo apt install curl
```

Get Latest Compatible ChromeDriver Version

```
LATEST_DRIVER=$(curl -s "https://googlechromelabs.github.io/chrome-for-testing/latest-patch-
```

```
versions-per-build.json" | jq -r ".builds[\"$CHROME_VERSION\"].version")
```

Download ChromeDriver

```
wget https://storage.googleapis.com/chrome-for-testing-public/$LATEST_DRIVER/linux64/  
chromedriver-linux64.zip
```

Extract ChromeDriver

```
unzip chromedriver-linux64.zip
```

Move ChromeDriver to System Path

```
sudo mv chromedriver-linux64/chromedriver /usr/local/bin/chromedriver  
sudo chmod +x /usr/local/bin/chromedriver
```

Verify ChromeDriver Installation

```
chromedriver --version
```

Example Output:

ChromeDriver 133.0.6943.141

The version should match your Chrome version.

Step 3: Add Selenium Dependency in pom.xml

Open the **pom.xml** file and add the Selenium dependency.

```
11  
12 <dependencies>  
13   <dependency>  
14     <groupId>junit</groupId>  
15     <artifactId>junit</artifactId>  
16     <version>3.8.1</version>  
17     <scope>test</scope>  
18   </dependency>  
19  
20  
21   <!-- https://mvnrepository.com/artifact/org.seleniumhq.selenium/selenium-java -->  
22   <dependency>  
23     <groupId>org.seleniumhq.selenium</groupId>  
24     <artifactId>selenium-java</artifactId>  
25     <version>4.29.0</version>  
26   </dependency>  
27 </dependencies>  
28
```

Step 4: Configure Maven Build Plugins

Add the following plugins in the **pom.xml** file.

```
29
30
31 <!-- Build: Configuring plugins and build settings -->
32 <build>
33 <plugins>
34 <!-- Example: Maven Compiler Plugin to compile Java code -->
35 <plugin>
36 <groupId>org.apache.maven.plugins</groupId>
37 <artifactId>maven-compiler-plugin</artifactId>
38 <version>3.8.1</version>
39 <configuration>
40 <source>1.7</source>
41 <target>1.7</target>
42 </configuration>
43 </plugin>
44 <!-- Example: Maven Surefire Plugin to run tests -->
45 <plugin>
46 <groupId>org.apache.maven.plugins</groupId>
47 <artifactId>maven-surefire-plugin</artifactId>
48 <version>2.22.2</version>
49 </plugin>
50 </plugins>
51 <plugin>
52   <groupId>org.apache.maven.plugins</groupId>
53   <artifactId>maven-jar-plugin</artifactId>
54   <version>3.0.1</version>
55   <configuration>
56     <archive>
57       <manifestEntries>
58         <Main-Class>com.example.App</Main-Class>
59       </manifestEntries>
60     </archive>
61   </configuration>
62 </plugin>
63 </build>
64 </project>
```

Step 5: Inspect Elements in the Website

Before writing Selenium code, we must inspect the HTML elements.

For Example :

Steps:

1. Open the website
<https://www.saucedemo.com/>
2. Right-click on the webpage.
3. Click **Inspect**.
4. Use the **cursor tool** to select product buttons.

Example attribute used for input fields:

Example:

Username : username

Password : password

Login Button : login-button

These help to login to the website.

Step 6: Create App.java

Location:

src/main/java/com/example/App.java

Java Code:

For : <https://www.saucedemo.com/>



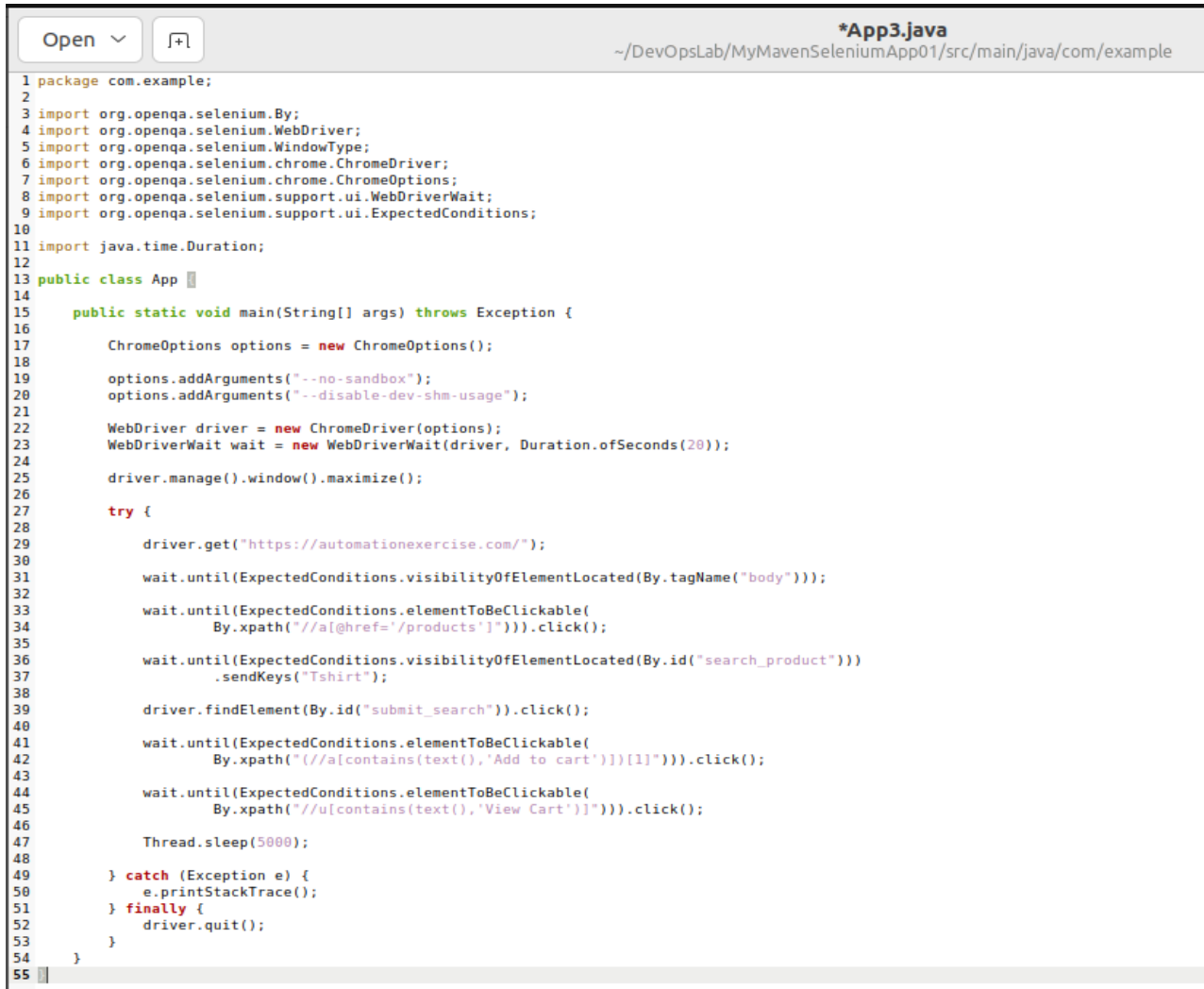
```
1 package com.example;
2
3 import org.openqa.selenium.By;
4 import org.openqa.selenium.WebDriver;
5 import org.openqa.selenium.chrome.ChromeDriver;
6
7 /**
8  * Hello world!
9  *
10 */
11 public class App
12 {
13     public static void main( String[] args ) throws InterruptedException
14     {
15         WebDriver driver=new ChromeDriver();
16         driver.get("https://practicetestautomation.com/practice-test-login/");
17
18         driver.manage().window().maximize();
19         Thread.sleep(2000);
20
21         driver.findElement(By.id("username")).sendKeys("student");
22         Thread.sleep(2000);
23
24         driver.findElement(By.id("password")).sendKeys("Password123");
25         Thread.sleep(2000);
26
27         driver.findElement(By.id("submit")).click();
28     }
29 }
```

For : <https://practicetestautomation.com/practice-test-login/>



```
1 package com.example;
2
3 import org.openqa.selenium.By;
4 import org.openqa.selenium.WebDriver;
5 import org.openqa.selenium.chrome.ChromeDriver;
6
7 /**
8  * Hello world!
9  *
10 */
11 public class App
12 {
13     public static void main( String[] args ) throws InterruptedException
14     {
15         WebDriver driver=new ChromeDriver();
16         driver.get("https://practicetestautomation.com/practice-test-login/");
17
18         driver.manage().window().maximize();
19         Thread.sleep(2000);
20
21         driver.findElement(By.id("username")).sendKeys("student");
22         Thread.sleep(2000);
23
24         driver.findElement(By.id("password")).sendKeys("Password123");
25         Thread.sleep(2000);
26
27         driver.findElement(By.id("submit")).click();
28     }
29 }
```


For : <https://automationexercise.com/>



```
1 package com.example;
2
3 import org.openqa.selenium.By;
4 import org.openqa.selenium.WebDriver;
5 import org.openqa.selenium.WindowType;
6 import org.openqa.selenium.chrome.ChromeDriver;
7 import org.openqa.selenium.chrome.ChromeOptions;
8 import org.openqa.selenium.support.ui.WebDriverWait;
9 import org.openqa.selenium.support.ui.ExpectedConditions;
10
11 import java.time.Duration;
12
13 public class App {
14
15     public static void main(String[] args) throws Exception {
16
17         ChromeOptions options = new ChromeOptions();
18
19         options.addArguments("--no-sandbox");
20         options.addArguments("--disable-dev-shm-usage");
21
22         WebDriver driver = new ChromeDriver(options);
23         WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(20));
24
25         driver.manage().window().maximize();
26
27         try {
28             driver.get("https://automationexercise.com/");
29
30             wait.until(ExpectedConditions.visibilityOfElementLocated(By.tagName("body")));
31
32             wait.until(ExpectedConditions.elementToBeClickable(
33                 By.xpath("//a[@href='/products']")))
34             .click();
35
36             wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("search_product")))
37             .sendKeys("Tshirt");
38
39             driver.findElement(By.id("submit_search")).click();
40
41             wait.until(ExpectedConditions.elementToBeClickable(
42                 By.xpath("//a[contains(text(),'Add to cart')][1]")))
43             .click();
44
45             wait.until(ExpectedConditions.elementToBeClickable(
46                 By.xpath("//u[contains(text(),'View Cart')]")))
47             .click();
48
49             Thread.sleep(5000);
50         } catch (Exception e) {
51             e.printStackTrace();
52         } finally {
53             driver.quit();
54         }
55     }
```

Step 7: Build the Maven Project

Run the following command:

```
mvn clean install
```

This command compiles the project and builds the JAR file.

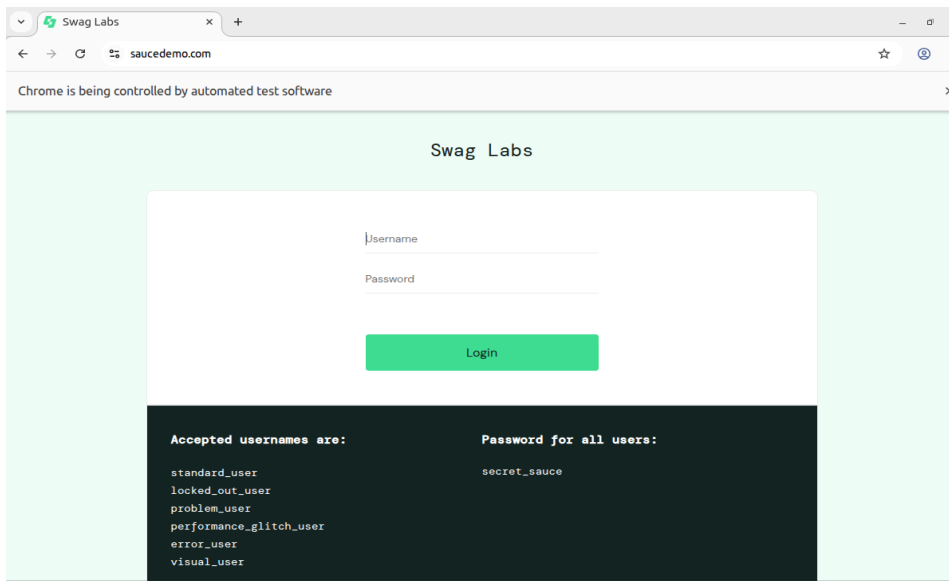
```
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyMavenSeleniumApp01$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:MyMavenSeleniumApp01 >-----
[INFO] Building MyMavenSeleniumApp01 1.0-SNAPSHOT
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MyMavenSeleniumApp01 ---
[INFO] Deleting /home/shravan-hegde/Devops/MyMavenSeleniumApp01/target
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenSeleniumApp01 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/shravan-hegde/Devops/MyMavenSeleniumApp01/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.11.0:compile (default-compile) @ MyMavenSeleniumApp01 ---
[INFO] Changes detected - recompiling the module! :source
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target/classes
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MyMavenSeleniumApp01 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/shravan-hegde/Devops/MyMavenSeleniumApp01/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.11.0:testCompile (default-testCompile) @ MyMavenSeleniumApp01 ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-surefire-plugin:2.12.4:test (default-test) @ MyMavenSeleniumApp01 ---
[INFO] No tests to run.
[INFO]
[INFO] --- maven-jar-plugin:2.4:jar (default-jar) @ MyMavenSeleniumApp01 ---
[INFO] Building jar: /home/shravan-hegde/Devops/MyMavenSeleniumApp01/target/MyMavenSeleniumApp01-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MyMavenSeleniumApp01 ---
[INFO] Installing /home/shravan-hegde/Devops/MyMavenSeleniumApp01/target/MyMavenSeleniumApp01-1.0-SNAPSHOT.jar to /home/shravan-hegde/.m2/repository/com/example/MyMavenSeleniumApp01/1.0-SNAPSHOT/MyMavenSeleniumApp01-1.0-SNAPSHOT.jar
```

java -jar target/MyMavenSeleniumApp01-1.0-SNAPSHOT.jar/mvn exec:java -Dexec.mainClass="com.example.App"

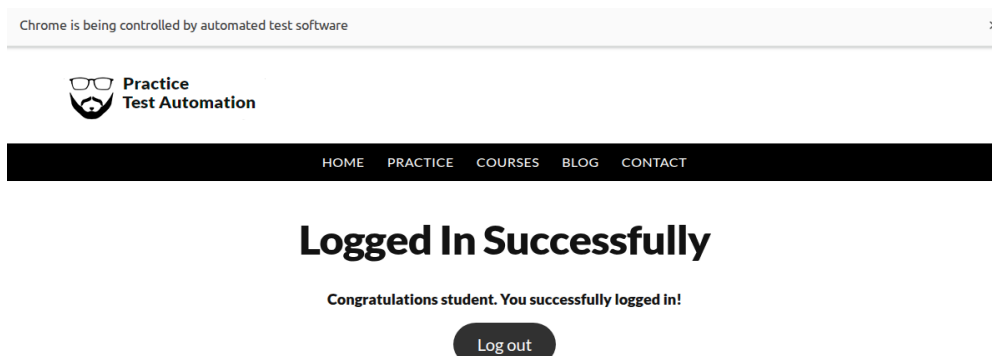
```
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MyMavenSeleniumApp01$ mvn exec:java -Dexec.mainClass="com.example.App"
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:MyMavenSeleniumApp01 >-----
[INFO] Building MyMavenSeleniumApp01 1.0-SNAPSHOT
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- exec-maven-plugin:3.1.0:java (default-cli) @ MyMavenSeleniumApp01 ---
Mar 18, 2026 1:14:35 AM org.openqa.selenium.devtools.CdpVersionFinder findNearestMatch
WARNING: Unable to find CDP implementation matching 145
Mar 18, 2026 1:14:35 AM org.openqa.selenium.chromium.ChromiumDriver lambda$new$4
WARNING: Unable to find version of CDP to use for 145.0.7632.109. You may need to include a dependency on a specific version of the CDP
using something similar to 'org.seleniumhq.selenium:selenium-devtools-v86:4.29.0' where the version ("v86") matches the version of the c
hromium-based browser you're using and the version number of the artifact is the same as Selenium's.
Logged in - current page title: Swag Labs
[WARNING] thread Thread[UrlChecker-2,5,com.example.App] was interrupted but is still alive after waiting at least 14994msecs
[WARNING] thread Thread[UrlChecker-2,5,com.example.App] will linger despite being asked to die via interruption
[WARNING] thread Thread[CompletableFutureDelayScheduler,5,com.example.App] will linger despite being asked to die via interruption
[WARNING] NOTE: 2 thread(s) did not finish despite being asked to die via interruption. This is not a problem with exec:java, it is a proble
m with the running code. Although not serious, it should be remedied.
[WARNING] Couldn't destroy threadgroup org.codehaus.mojo.exec.ExecJavaMojo$IsolatedThreadGroup[name=com.example.App,maxpri=10]
```

Outputs :

For : <https://www.saucedemo.com/>



For : <https://practicetestautomation.com/practice-test-login/>



For : <https://automationexercise.com/>

